KSHITIZ RAJ PAUDYAL

kshitizrajpaudyal@gmail.com

Kathmandu, Nepal | kshitizrajpaudyal.com.np | github.com/Kshitiz726

EDUCATION

Tribhuvan University, Institute of Engineering, Thapathali Campus, Kathmandu, Nepal 2022-2026

Bachelor of Engineering in Computer Engineering **Aggregate: 76.32%**

- **Relevant coursework:** Data Mining, Big Data Technologies, Artificial Intelligence, Probability and Statistics, Simulation and Modelling, Numerical Methods, Digital Signal Analysis and Processing, Multimedia System

RESEARCH INTERESTS

Spatiotemporal Deep Learning, Time-Series Forecasting, Multimodal Learning (Vision, Audio and Language), Video Understanding, Machine Learning for Environmental Data

PUBLICATIONS AND RESEARCH

- **Paudyal, K. R.**, Shakya, R., & Upadhayaya, J. (2025). Spatiotemporal PM2.5 Estimation in Kathmandu Using Deep Learning with OpenMeteo and NASA MERRA-2 Data: Performance Benchmarking Against Machine Learning Model. *International Journal on Engineering Technology (InJET)*, 3(1), 36-45. Peer-reviewed. DOI: 10.3126/injet.v3i1.87012 **(First and corresponding author)**
  Built a deep neural network from weather and satellite reanalysis data to reconstruct hourly air pollution; reached R^2 = 0.87 and benchmarked it against XGBoost.
- **Paudyal, K. R.**, Tamang, A., Shah, N., Bhattarai, S., & Shrestha, A. A Multimodal AI System for Ranking and Extracting Highlight Clips from Long-form Videos Using Visual, Audio, and Textual Fusion. *Journal of Innovations in Engineering Education (JIEE)*. **(First author, under peer review)**
  Final year project. Fuses visual, speech and audio signals (BERT-based) to score, rank and extract key clips automatically.

PRESENTATIONS

- Presenter, 4th Thapathali Graduate Conference, IOE Thapathali Campus (2026)

AWARDS AND HONOURS

- **Best Final Year Project**, selected by the Department Project Committee, IOE Thapathali Campus (2026)
- **Licensed Computer Engineer**, registered with the Nepal Engineering Council

WORK EXPERIENCE

Freelance Software Developer (Remote) Oct 2025-Present

The Social Circle (The Social Globe Ltd), United Kingdom

- Design, develop and maintain websites, including their back-end systems, for The Social Circle and the brands it manages
- Develop the OurMynd™ iOS application (ourmynd.com), an ADHD productivity app, including its AI coach and focus rooms, using Flutter, Node.js, TypeScript, PostgreSQL and Redis

COURSES AND CERTIFICATIONS

- Machine Learning Specialization, University of Washington (Coursera)
- CS50AI: Introduction to Artificial Intelligence with Python, Harvard University

LEADERSHIP

Organizer, Yathartha Tech Fest, IOE Thapathali Campus 2024

- Helped plan and run technical events, managing schedules and event-day coordination

Campus Ambassador, LOCUS 2025, National Technological Exhibition, IOE Pulchowk 2025

- Promoted the exhibition and its tech competitions to students at Thapathali Campus

SKILLS

- Machine Learning and Data Science: Python, TensorFlow, XGBoost, BERT, OpenCV
- Data and Systems: SQL, PostgreSQL, Redis, Node.js, TypeScript, Flutter, Git
- English Proficiency: IELTS, Overall Band 7.0 (July 2026)